

# Statistics 154 / 254: Statistical machine learning

UC Berkeley, Fall 2026

## Instructors



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### ! Important

If you are a concurrent enrollment student hoping to enroll in the class, I hope you will be able to join us!

In the meantime, please join BCourses and hand in your assignments as if you were enrolled. We will not retroactively waive assignments that were due prior to being enrolled in the class. Preference will be given to students who are demonstrably participating in the class.

Please reach out to [Erez](#) if you need to be manually added to BCourses.

This website will contain lecture materials and assignments. Day-to-day announcements and discussion will be found in ED. (See links above.)

## Materials

The course will be based largely on [Lecture notes](#). The notes can be supplemented using subsets of the following texts:

- [Patterns, predictions, and actions: A story about machine learning](#) Hardt, Recht
- [ESL: The Elements of Statistical Learning](#) Hastie, Tibshirani, Friedman
- [ISL: An Introduction to Statistical Learning](#) James, Witten, Hastie, Tibshirani

Additional reading will supplement these texts as necessary.

Some other good sources of reading material are:

- [Understanding Machine Learning: From Theory to Algorithms](#) Shalev-Shwartz, Ben-David
- [Probabilistic Machine Learning: An Introduction](#) Murphy
- [Dive into Deep Learning](#) Many authors

## Schedule

Lectures will be held Aug 27th 2026 – Dec 3rd 2026 from 3:30pm – 5pm (starting at Berkeley time, 3:40pm) in Valley Life Sciences 2060.

Labs will be held on Mondays from

- 9am–10am (Gateway B1026)
- 4pm–6pm (Gateway B1012)
- 11am–1pm (Gateway B1023)
- 1pm–3pm (Gateway B1023)

The [official course catalog entry](#) is the ground truth for scheduling. If the catalog conflicts with this webpage, please notify the instructor and trust the catalog.

The tentative week-by-week course calendar is as follows. **The following schedule is aspirational and subject to change as we go.**

According to the [official schedule](#), the final exam has not been scheduled yet .

Table 1: Calendar (tentative)

| Date   | Day            | Note                           | Unit                               | Topic | Assign-<br>ment  |
|--------|----------------|--------------------------------|------------------------------------|-------|------------------|
| Aug 27 | Thurs-<br>day  | Lecture                        | Course policies                    |       |                  |
| Aug 31 | Monday         | Lab                            |                                    |       | Python<br>review |
| Sep 1  | Tuesday        | Lecture                        | Unit 0: Introduction<br>and review |       |                  |
| Sep 3  | Thurs-<br>day  | Lecture                        |                                    |       | HW 0             |
| Sep 7  | Monday         | Adminis-<br>trative<br>holiday | Unit 1: Regression                 |       |                  |
| Sep 8  | Tuesday        | Lecture                        |                                    |       |                  |
| Sep 10 | Thurs-<br>day  | Lecture                        |                                    |       | Quiz 0           |
| Sep 14 | Monday         | Lab                            |                                    |       | Chal-<br>lenge 1 |
| Sep 15 | Tuesday        | Lecture                        |                                    |       |                  |
| Sep 16 | Wednes-<br>day | Add /<br>drop<br>deadline      |                                    |       |                  |

| Date   | Day      | Note    | Unit                            | Topic | Assignment  |
|--------|----------|---------|---------------------------------|-------|-------------|
| Sep 17 | Thursday | Lecture | Unit 2: Classification          |       |             |
| Sep 21 | Monday   | Lab     |                                 |       | Challenge 1 |
| Sep 22 | Tuesday  | Lecture |                                 |       |             |
| Sep 24 | Thursday | Lecture |                                 |       | HW 1        |
| Sep 28 | Monday   | Lab     |                                 |       | Challenge 1 |
| Sep 29 | Tuesday  | Lecture | Unit 3: Risk and Complexity     |       |             |
| Oct 1  | Thursday | Lecture |                                 |       | Quiz 1      |
| Oct 5  | Monday   | Lab     |                                 |       | Reading 1   |
| Oct 6  | Tuesday  | Lecture |                                 |       |             |
| Oct 8  | Thursday | Lecture |                                 |       | HW 2        |
| Oct 12 | Monday   | Lab     | Unit 4: Trees and weak learners |       | Challenge 2 |
| Oct 13 | Tuesday  | Lecture |                                 |       |             |
| Oct 15 | Thursday | Lecture |                                 |       | Quiz 2      |
| Oct 19 | Monday   | Lab     |                                 |       | Challenge 2 |
| Oct 20 | Tuesday  | Lecture |                                 |       |             |
| Oct 22 | Thursday | Lecture | Unit 4: Trees and weak learners |       |             |
| Oct 26 | Monday   | Lab     |                                 |       | Challenge 2 |
| Oct 27 | Tuesday  | Lecture |                                 |       |             |
| Oct 29 | Thursday | Lecture |                                 |       | HW 3        |
| Nov 2  | Monday   | Lab     |                                 |       | Reading 2   |
| Nov 3  | Tuesday  | Lecture |                                 |       |             |
| Nov 5  | Thursday | Lecture |                                 |       | Quiz 3      |

| Date   | Day       | Note                   | Unit                                     | Topic | Assignment  |
|--------|-----------|------------------------|------------------------------------------|-------|-------------|
| Nov 9  | Monday    | Lab                    |                                          |       | Challenge 3 |
| Nov 10 | Tuesday   | Lecture                |                                          |       |             |
| Nov 11 | Wednesday | Administrative holiday |                                          |       |             |
| Nov 12 | Thursday  | Lecture                |                                          |       |             |
| Nov 16 | Monday    | Lab                    |                                          |       | Challenge 3 |
| Nov 17 | Tuesday   | Lecture                | Unit 5: Neural networks and optimization |       |             |
| Nov 19 | Thursday  | Lecture                |                                          |       | HW 4        |
| Nov 23 | Monday    | Lab                    |                                          |       | Challenge 3 |
| Nov 24 | Tuesday   | Lecture                |                                          |       |             |
| Nov 25 | Wednesday | Thanksgiving           |                                          |       |             |
| Nov 26 | Thursday  | Thanksgiving           |                                          |       |             |
| Nov 27 | Friday    | Thanksgiving           |                                          |       |             |
| Nov 30 | Monday    | Lab                    |                                          |       | Reading 3   |
| Dec 1  | Tuesday   | Lecture                |                                          |       | Quiz 4      |
| Dec 3  | Thursday  | Lecture                | Class review                             |       | HW 5        |